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The Impact of Momentum on Tennis Scoring

Summary

Momentum usually refers to a player's performance and form during a match, as well as the advantage or momentum they gradually gain during a match.

In Task 1, firstly, we perform a data preprocessing operation on the original data. Then, select relevant indicators from the preprocessed data to establish **Player Performance Evaluation Model**, including winning a point on a serve, consecutive score, breaking serve score and forced error. The typical Carlos Alcaraz and Novak Djokovic matches are then selected for visualization and analysis, showing the average performance of the players at each point, game and set, respectively. The images illustrate that a player's momentum can reflect the player's scoring to a certain extent.

In Task 2, **Spearman Correlation** is used to study the correlation between player momentum and scoring. The absolute value of the correlation coefficient is **0.89**, which means there is a strong correlation. Furthermore, a randomness test is performed to test the relationship between momentum and score. Selecting the match of Carlos Alcaraz and Novak Djokovic, randomize the score 1000 times, and in each instance calculate the momentum by using the Model in Task 1. Finally, the momentum mean is taken as the "random" momentum fluctuation. The actual and randomized momentum volatility series are then tested for differences using the **Mann-Whitney U-test**. The results indicate that there is strong variability between the two series, thus, further suggesting there is a correlation between momentum fluctuations and scores.

In Task 3, we define the **Player Advantage Characteristic** reaction momentum fluctuations. Next, relevant indicators are selected, mainly including Scoring Feature, Rate Feature and Activity Feature. After that, Logistic Regression, Decision Tree, **Random Forest**, KNN and SVM models are selected as base models for training and prediction, respectively. The results show that the Random Forest model has the optimal prediction results with **0.69** for all evaluation indicators. Therefore we construct **Match Advantage Prediction Model** based on Random Forest Model. Based on the feature importance scores of the Random Forest Model, we derive the variables that are most relevant to player advantage. Finally, based on the above results, we can provide players with recommendations.

In Task 4, we collect relevant data from several website in order to test established Match Advantage Prediction Model. We use the additional 2023 Wimbledon Gentlemen Championships data collected as a test set, and the results show that all evaluation metrics are above **0.6**, which is an accurate prediction of player advantage. In addition, we use the Women's matches, US Open matches and Table tennis matches data inputs respectively into the Match Advantage Prediction Model to predict player advantage, and the output shows that neither gender nor tennis tournament type had an effect on the applicability of the model. For different ball sports, due to the lack of corresponding indicators, we can only build simple models with poor prediction results. Subsequently, if we can add feature data in the future, we will be able to make better predictions.

A report that mainly includes the results of the research conducted by this paper is written, while giving suggestions for coaches and players to deal with the momentum of the match, respectively.

Keywords: Player Performance Evaluation Model, Match Advantage Prediction Model,

Momentum, Tennis

Content

1 Introduction 4
1.1 Background 4
1.2 Restatement 4
1.3 Our Work 5
2 Assumptions and Justifications 5
3 Notations 5
4 Model I: Player Performance Evaluation Model 6
4.1 Introduction to Relevant Background 6
4.1.1 Introduction to the rules of tennis events 6
4.1.2 What is momentum? 6
4.2 Data Processing 7
4.2.1 Missing value handling 7
4.2.2 Data type conversion 7
4.3 Model Establishment 7
4.4 Result of Model I 8
4.5 Correlation Analysis Between Momentum and Score 9
4.5.1 Correlation analysis of momentum and score 9
4.5.2 Randomness test of momentum and score 10
5 Model II: Match Advantage Prediction Model 12
5.1 What is Advantage Shift in Momentum? 12
5.2 Establish Match Advantage Prediction Model 12
5.2.1 Model feature selection 12
5.2.2 Relevant model selection 13
5.3 Random Forests Model in Details 13
5.4 Match Advantage Prediction Model Results 14
5.5 Analysis of The Most Relevant Factors 15
5.5.1 Random Forest feature importance score 15
5.5.2 Visual presentation of the most relevant factors 15
5.6 Advice on A Player Going into A New Match against A Different Player 16
5.7 Test and Application on Match Advantage Prediction Model 17
5.7.1 Test on Match Advantage Prediction Model 17
5.7.2 Application on Match Advantage Prediction Model 17

6 Sensitivity Analysis 20

7 Strengths and Weaknesses 20

7.1 Strengths 20

7.2 Weaknesses 20

References 21

Report on Use of AI 22

Appendix 23

1 Introduction

1.1 Background

Recently, as technology and data analysis develop by leaps and bounds, researches concerning about momentum and situation changes in athletic competition deepened and have become more comprehensive. Momentum, regarded as one of the significant factors in the competition, is able to change the tendency of the competition in a short period of time and is a powerful force in players' performance. With a view to understanding and predicting the changes in momentum precisely, this paper takes various of factors that contributed to the outcome of the match into account and develop appropriate mathematical models for analysis and inference. By delving into the laws of momentum and situation change, we can offer coaches, players and audiences an in-depth athletics tournament analysis and decision support, driving the evolution and innovation of athletics tournament.



Figure 1.1 tennis and tennis racket picture

1.2 Restatement

- Develop a mathematical model, considering comprehensively the pertinent factors in players. This model can explain which player better acquits himself and quantifies his advantage in performance in a certain moment during the match. Using visual tables and figures to describe the trend of the match.
- Evaluate the effect of the momentum on players' success. Analyze whether there is a certain relationship or randomness between momentum and players' success.
- Select the indicators from the match data to build a model and predict fluctuations in player advantage during matches. Find out the most relevant factors and provide suggestions for players based on these factors.
- Testing the model established above with data from the remaining matches, while considering the factors that need to be added if the model underperforms. Then apply the model to woman's competitions, other tennis competitions and even other sports competitions so as to determine its applicability.

1.3 Our Work

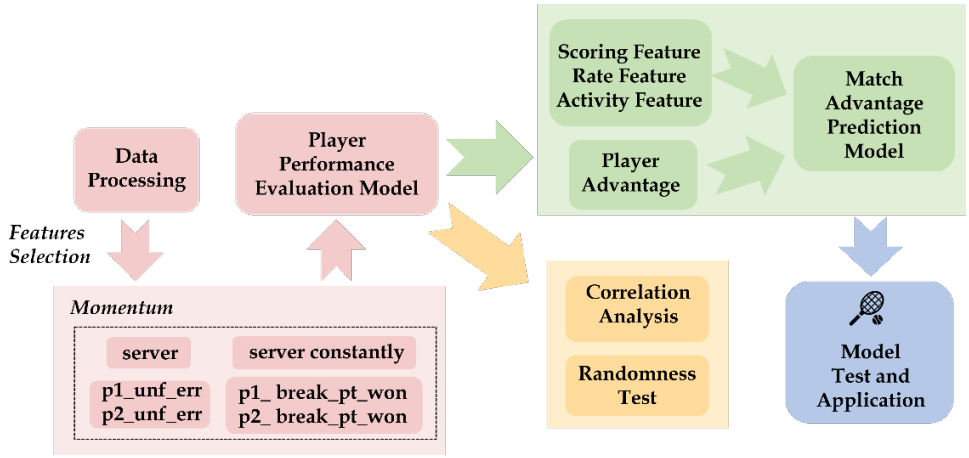


Figure 1.2 Our work

From Figure 1.2, different colors represent different tasks, pink blocks represent task 1, orange blocks represent task 2, green blocks represent task 3, blue represents task 4.

2 Assumptions and Justifications

To simplify the analysis of our problem, we make the following assumptions, each of which is properly justified.

- Assume that players are able to fully recovered after break, i.e., the movement level of the players only indicates how active they are.
- Assume that physical fitness levels of the different players in each game have no effect on player performance.
- Assume that the conditions of the stadium facilities are the same for each match, which means, the stadium environment has no effect on the performance of the players.
- Assume that the verbal and adjudicative behavior of the referees and other staff during the match does not have an impact on player performance.

3 Notations

The primary notations used in this paper are listed in Table 3.1.

Table 3.1 Notations

Symbol	Description
P_{1t}	The performance of player 1 at moment t
w_{1t}	The weight of player1's score at moment t
a_{1t}	Whether player 1 win the point by breaking serve at t moment
b_{1t}	Whether player 1 win the point by forced error at t moment
α	The impact factor of w_{1t}
β_1	The impact factor coefficient of a_{1t}
β_2	The impact factor coefficient of b_{1t}
D_t	Player advantage in the current situation at moment t

4 Model I: Player Performance Evaluation Model

4.1 Introduction to Relevant Background

4.1.1 Introduction to the rules of tennis events

Tennis is divided into points, games, sets and matches. Points form a game. Games form a set. And sets form a match. Points are subdivided into 0, 1, 2, 3, and 4 denoted by 15, 30, and 40 respectively in the game. The winning conditions for each game are to win four points and have at least two more points than your opponent, and if a deuce draw is reached, you need to score two consecutive points. The winning conditions for each set are to win 6 games and win at least two more sets than the opponent. A tie-break is required when 6:6 is reached. Win the match by winning 5 sets, cf. Figure 4.1.

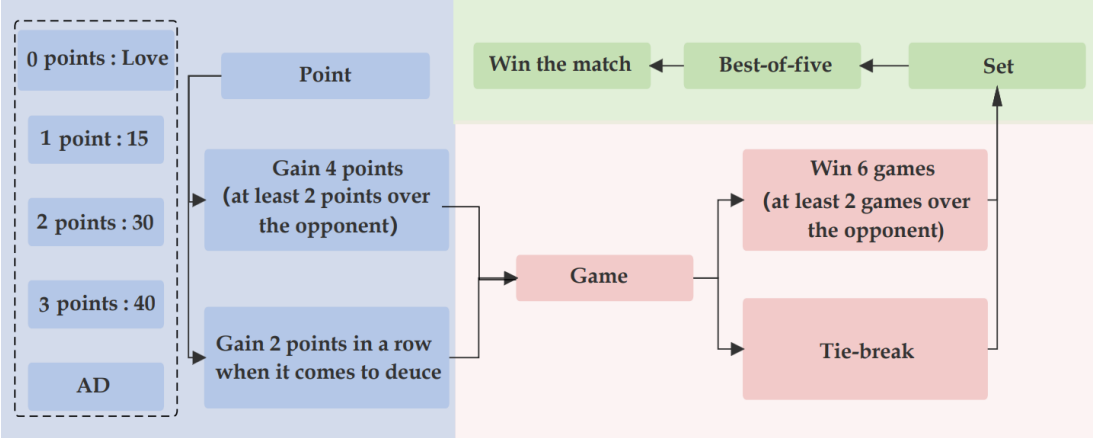


Figure 4.1 Tennis rules flow chart

4.1.2 What is momentum?

Momentum usually refers to a player's performance and form during a match, as well as the advantage or momentum they gradually gain during a match. Momentum means that a player benefits from a psychological or physiological boost. Momentum usually includes the continuity of victory, the consistency of performance, the defeat of opponents, the state of mind. What's more, momentum is probably influenced by exercise status and physical conditions, technical and tactical adjustments, the strength of adversaries, and external factors, cf. Figure 4.2.

Folk psychology suggests numerous (and often contradictory) ways in which relative positions in competition may affect performance[7]. In a seminal paper[7], Ferrall and Smith (1999), derived a Nash equilibrium for the problem to study "best of 7 games" in the final series in Basketball, Baseball and Hockey in the United States. Klumpp and Polborn(2006) used this framework to model the sequential race to the primaries and Konrad and Kovenock (2008) proposed a theoretical analysis of best-of-n competition when each stage is modelled as an all-pay auction. The theoretical results of these papers indicate that a contestant in the lead will be more likely to have an interest in expending more effort than a trailing contestant.

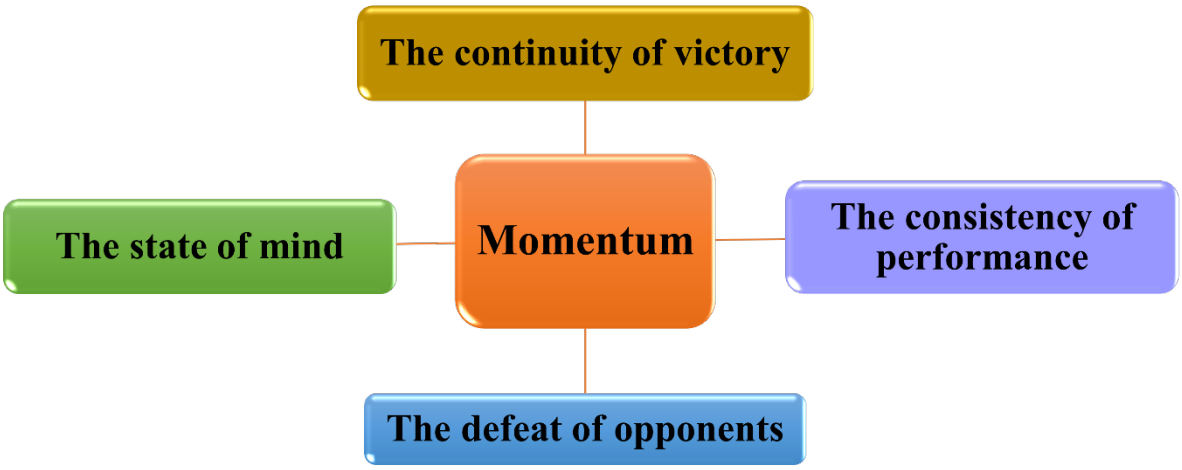


Figure 4.2 Momentum Influencing factor diagram

4.2 Data Processing

4.2.1 Missing value handling

In the given data, there are missing values in the variables speed_mph, serve_width, serve_depth and return_depth. Speed_mph is a numeric variable. Therefore, we adopt a mean-padding approach. By virtue of serve_width,serve_depth and return_depth are string variables, plurality filling method is used.

4.2.2 Data type conversion

In the given data, the feature of p1_score and p2_score exist the value AD, which is non-numeric. Consequently, we convert AD to 50 for facilitating subsequent model processing.

4.3 Model Establishment

It is assumed that the performance of two players is P_{1t} and P_{2t}. Among them, t represents players performance in a certain moment. Initially, two players performances are both 0. According to the relevant background, we suppose that a player who performs better, i.e.. higher performance value. The players' performance equation is defined as follows.

$$P_{1t} = \begin{cases} P_{1t-1} + 1 & if\ player\ 1\ win\ the\ point\ at\ t\ moment \\ 0 & else \end{cases} \quad (1)$$

For player 2, the performance equation is similar, so there is unnecessary to go into details.

In the tennis match, server is much more likely to win the point. It is needed to revise the above equation to calculate the performance of players. The details are as follows.

$$P_{1t} = \begin{cases} (P_{1t-1} + 1) \times w_{1t} & if\ player\ 1\ win\ the\ point\ at\ t\ moment \\ 0 & else \end{cases}, \quad (2)$$

where w_{1t} represents the weight of player 1's score at the very moment. The specific equation is shown as follows.

$$w_{1t} = \begin{cases} 1 - \alpha & if\ player\ 1\ win\ the\ point\ on\ a\ serve\ at\ t\ moment \\ 1 + \alpha & if\ player\ 1\ win\ the\ point\ on\ a\ return\ at\ t\ moment, \\ 1 & else \end{cases} \quad (3)$$

where α is the impact factor of w_{1t} and is set to be 0.2.

Taking the effect of consecutive scoring on player's performance into consideration, a player who consecutively wins the point performs better. The equation (3) is modified.

$$P_{1t} = \begin{cases} (P_{1t-1}+1)\times w_{1t} + C & if\ player\ 1\ win\ three\ or\ more\ points\ at\ t\ moment \\ (P_{1t-1}+1)\times w_{1t} & if\ player\ 1\ win\ the\ point\ at\ t\ moment \\ 0 & else \end{cases}, \quad (4)$$

where C is a constant and should be added when a player wins more than three points consecutively. C is assigned to 5.

During the match, players usually will be influenced by psychological condition. Under the situation, it is believed that breaking serve score and forced errors have a greater impact on a player's mental state. Therefore, the above equation is modified again. a_{1t} means whether player 1 win the point by breaking serve at t moment, b_{1t} means whether player 1 win the point by forced error at t moment. If yes, the value is assigned to 1. If not, the value is assigned to 0. β_1 and β_2 are impact factor coefficient of a_{1t} and b_{1t} respectively. The final performance equation for player 1 is as follows.

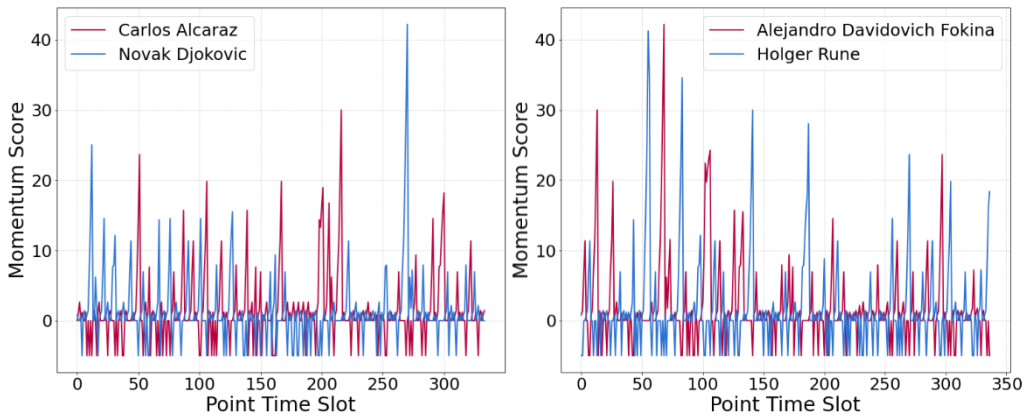
$$x = a_{1t}\beta_1 - b_{1t}\beta_2 \quad (5)$$

$$P_{1t} = \begin{cases} (P_{1t-1}+1)\times w_{1t} + x + C & if\ player\ 1\ win\ three\ or\ more\ points\ at\ t\ moment \\ (P_{1t-1}+1)\times w_{1t} + x & if\ player\ 1\ win\ the\ point\ at\ t\ moment \\ x & else \end{cases}, \quad (6)$$

where the value of β_1 and β_2 is set to 5.

4.4 Result of Model I

The equation (6) is used to calculate the preprocessed data. Different players performance in every match can be obtained. In order to illustrate the result, we choose two matches with match_id "2023-wimbledon-1701" and "2023-wimbledon-1304" to visualize the performance of different players. We visualize each point, each game and each set from the perspective of the players, shown as below.



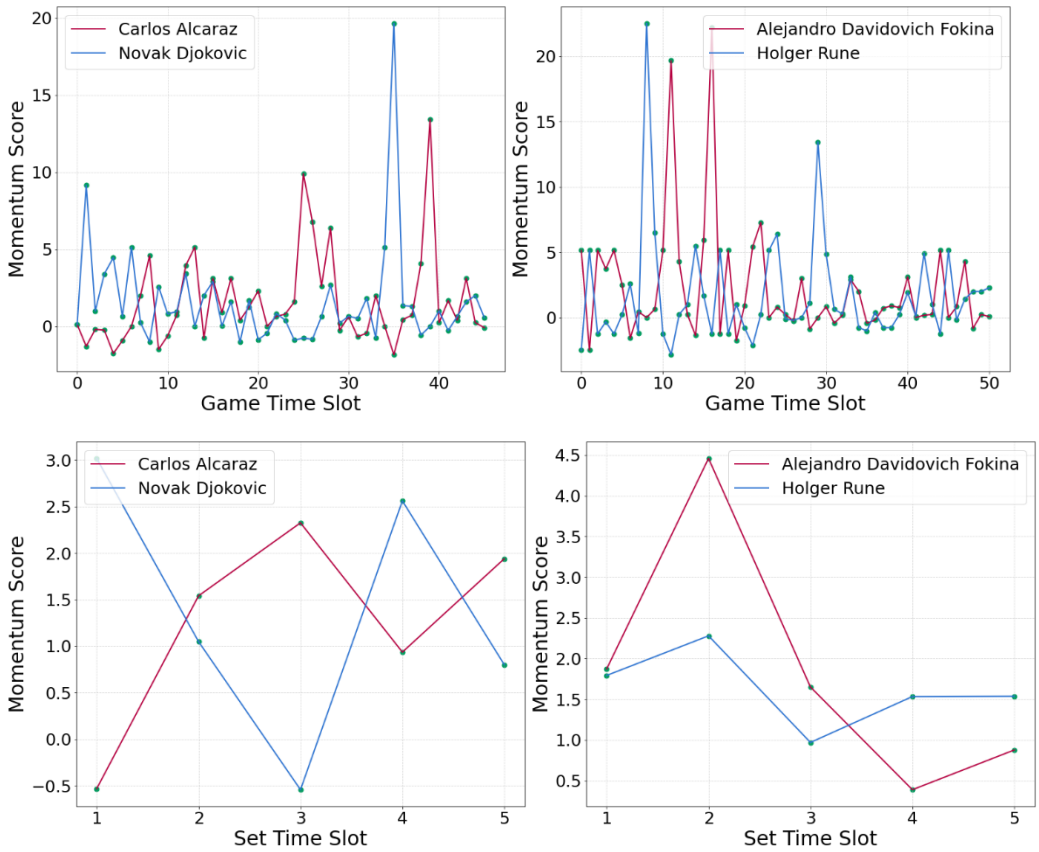


Figure 4.3 The performance scores from the points, games and sets of the match

From Figure 4.3, from the view of points, it is difficult to distinguish which player is better. In contrast, from the view of games, we can better judge the performance of the players. It is clear to find that over a continuous period of time, there will be situations in which one player's performance advantage is greater than that of another player. From the view of sets, the player's performance in each set is exactly representative of the number of sets won by the player in the current situation.

Specifically, we take the typical match between Carlos Alcaraz and Novak Djokovic as an example (on the left side of the picture above). It is obvious to find that the combined performance points of each set between Carlos Alcaraz and Novak Djokovic exactly reflect the current situation. In the first set, Novak Djokovic showed a huge margin of victory over Carlos Alcaraz. In the second set, Carlos Alcaraz did perform slightly better than Novak Djokovic. In the third set, Carlos Alcaraz had the dominant performance over Novak Djokovic. But in the fourth set, Novak Djokovic again took control, and he had the advantage over Carlos Alcaraz. However, in the final set, Carlos Alcaraz had a bigger advantage and he won the match. In each set, we break it down into games and then into points. As a result, the "momentum" of the players can better explain the result of the match.

4.5 Correlation Analysis Between Momentum and Score

4.5.1 Correlation analysis of momentum and score

A normality test is run for the momentum in each match. We choose Shapiro-Wilk for the normality test, where p1_momentum and p2_momentum represent the momentum of player 1

and player 2, respectively. The normality test results are shown as below.

Table 4.1 The results of the momentum normality test for each match

Feature	Statistic W	P_value
p1_momentum	0.60531	6.207e-27
P2_momentum	0.53384	8.362e-29

From Table 4.1, for both p1_momentum and p2_momentum, the p-value is less than 0.05, i.e., there is a significant difference, so it can be concluded that for both p1_momentum and p2_momentum, none of them conform to the normal distribution.

For data that do not conform to the normal distribution, the spearman correlation coefficient is applied to analyze the correlation between momentum and point, and the correlation coefficient results are shown as follows.

Table 4.2 The correlation coefficient between momentum and score

	point_victor	p1_momentum	P2_momentum
point_victor	1	-0.89	0.89
p1_momentum	-0.89	1	-0.83
P2_momentum	0.89	-0.83	1

From Table 4.2, the absolute value of correlation coefficient p1_momentum, p2_momentum and point_victor is 0.89. It shows that the correlation between momentum and point is strong, indicating that there is a certain correlation between momentum fluctuation and success of a player in a match. It is not random, cf. Figure 4.4.



Figure 4.4 Correlation Analysis Between Momentum and Score

4.5.2 Randomness test of momentum and score

To determine whether the relationship between momentum fluctuations and points is random, we propose randomness test as follows.

Step 1: Select a match, randomly assign the score for that match.

Step 2: Calculate player momentum based on the player performance equation.

Step 3: Repeat Step 1, 2 multiple times, averaging the resulting momentum. Then subject to a difference test on randomized momentum series and the original momentum series.

We choose the typical match between Carlos Alcaraz and Novak Djokovic which the-match_id is "2023-wimbledon-1701" to test the randomness. Since the randomized points conflict with the player's true breaking serve score or forced error, we use equation (4) to calculate the player's momentum. Meanwhile, we choose the number of repetitions to be 1000 times. In this way the error caused by randomness can be avoided. The final results are shown as below.

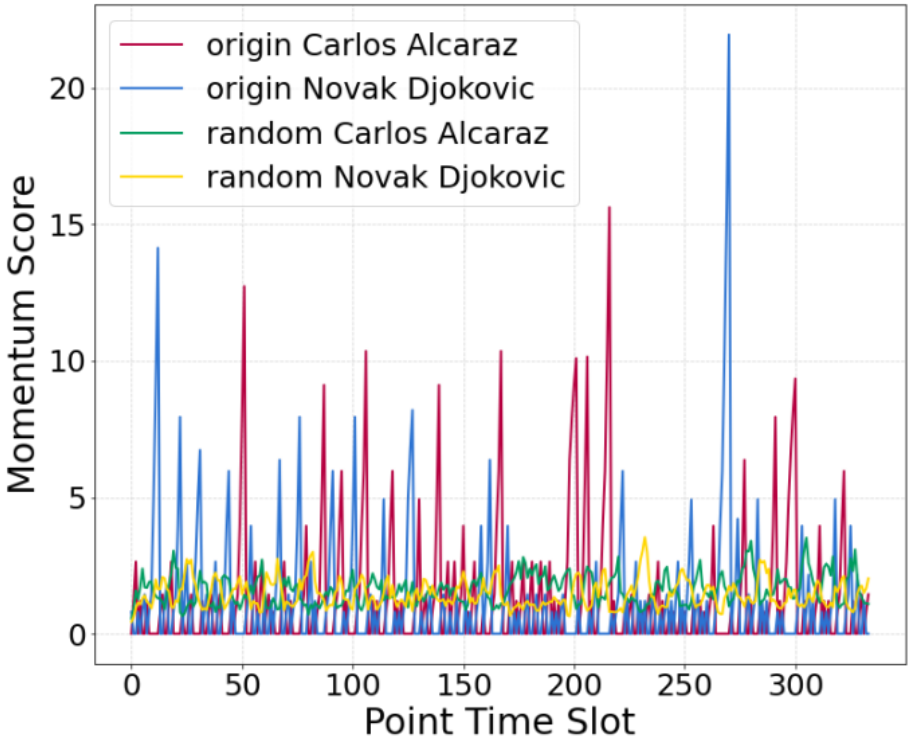


Figure 4.5 Momentum fluctuation of players

From Figure 4.5, it indicates that the players' momentum changes after randomness test. The players' momentum changes in the original match have a significant difference, which proves that the momentum and the point are not randomly distributed. In order to determine more scientifically, we use non-parametric tests to verify. In this case, we choose the Mann-Whitney U-test to test whether there is a discrepancy between the two sets of sequences. The final results are shown as follows.

Table 4.3 Results of variability test of the momentum series

Feature	Statistic	P_value
p1_momentum	26647.0	5.246e-32
P2_momentum	27049.0	3.381e-31

From Table 4.3, for both p1_momentum and p2_momentum, the p-value is less than 0.05, i.e., there is a significant difference. The significant variability between the two sets of data provides further evidence that there is a correlation, rather than randomness, between a player's momentum fluctuations and game success.

5 Model Ⅱ: Match Advantage Prediction Model

5.1 What is Advantage Shift in Momentum?

We should judge when the flow of play is about to change from favoring one player to the other. It is difficult for us to determine the turning point just from the change of momentum. In comparison, the advantage of a player can be defined. We define the advantage of players in current situation as D_t.

$$D_t = \begin{cases} 1 & condition1 \\ 2 & condition2, \\ 0 & else \end{cases} \tag{7}$$

where condition 1 is the momentum of player 1 continuously exceeds that of player 2 for 3 points or more at t moment. Condition 2 is the momentum of player 2 continuously exceeds that of player 1 for 3 points or more at t moment.

For the equation (7), when the momentum of player 1 is ahead of player 2 in three consecutive points at t moment. It means that player 1 has performed well for a period of time, indicating that he has an advantage over player 2, and vice versa. Otherwise, we believe that the momentum between player 1 and player 2 fluctuates greatly, indicating that the current situation is an even match.

5.2 Establish Match Advantage Prediction Model

5.2.1 Model features selection

Features like p1_sets, p2_sets, p1_games, p2_games, p1_score, p2_score, p1_distance_run,p2_distance_run, rally_count, speed_mph are selected. These primitive features are directly used as input features to the model. Because these features all represent the current situation. Features like p1_ace which indicates whether player 1 made the winning shot in the current game value 0 or 1. We suppose that these data features are unable to represent the current status of players. Therefore, we turn it into calculating the percentage of winning shots issued in the current game, and set to p1_ace_rate. We process the remaining variables as input to the model above. The processing results are shown in the following table.

Table 5.1 The input features

Variables	Transformed variables	Variables	Transformed variables
p2_ace	p2_ace_rate	p1_unf_err	p1_unf_err_rate
p1_winner	p1_winner_rate	p2_unf_err	p2_unf_err_rate
p2_winner	p2_winner_rate	p1_net_pt	p1_net_pt_rate
p1_doble_fault	p1_doble_fault_rate	p2_net_pt	p2_net_pt_rate
p2_doble_fault	p2_doble_fault_rate	p1_net_pt_won	p1_net_pt_won_rate
p1_break_pt	p1_break_pt_rate	p2_net_pt_won	p2_net_pt_won_rate
p2_break_pt	p2_break_pt_rate	p1_break_pt_missed	p1_break_pt_missed_rate
p1_break_pt_won	p1_break_pt_won_rate	p2_break_pt_missed	P2_break_pt_missed_rate
p2_break_pt_won	p2_break_pt_won_rate		

5.2.2 Relevant model selection

For the traditional model, we choose the Logistic Regression. In addition, we choose Decision Tree, Random Forest, KNN, SVM machine learning models.

Logistic Regression is a linear model used for classification problems. Decision Tree make decisions by means of branches and nodes. SVM is a powerful model for classification and regression that separates samples of different classes by finding an optimal hyperplane. Random Forest is an ensemble learning method, which consists of multiple Decision Tree. KNN is a classification algorithm based on proximity samples, which makes predictions by calculating the distance between samples.

We apply the above models, using the data features as the input to the model and the player advantage as the dependent variable. Machine learning model requires a large amount of data, so we use all the provided data and divide it into training set and test set according to the ratio of 8.5:1.5. At the same time, the parameters of the model were optimized to obtain the optimal fitting effect of each model, as shown below.

Table 5.2 The optimal fitting effect of each model

Model	accuracy	precision	recall	f1-score
Logistics Regression	0.50	0.51	0.50	0.49
Decision Tree	0.60	0.61	0.60	0.60
Random Forest	0.69	0.69	0.69	0.69
KNN	0.49	0.49	0.49	0.48
SVM	0.48	0.52	0.48	0.41

From the Table 5.2, each evaluation index of decision tree and Random Forest model is better than other models. The possible reasons are as follows. Decision Tree and Random Forests can deal with high dimensional data efficiently. Because every feature can be used to construct branch condition of decision tree. Under the condition of massive features, Decision Tree can be used for feature selection and segmentation more comprehensively. At the same time, Random Forest combines the prediction results of multiple decision tree models to reduce the variance and overfitting risk. So it can improve the overall classification performance compared with the decision tree model. Based on the above results, we choose Random Forest as the base model for prediction.

5.3 Random Forests Model in Details

Random Forest is an integrated learning method based on Bagging that can be used for problems such as classification and regression. It has many advantages, such as high accuracy, strong resistance to overfitting, applicability to high-dimensional data and easy parallelization.

The data prediction process of Random Forest includes the following steps.

1) Randomly selecting sample data from the basic dataset to form n different sample datasets.
2) Constructing decision tree models for each sample data set, with a total of n different decision tree models.
3) Using the average of these decision tree models to obtain the final prediction results.

The specific process is shown in Figure 5.1.

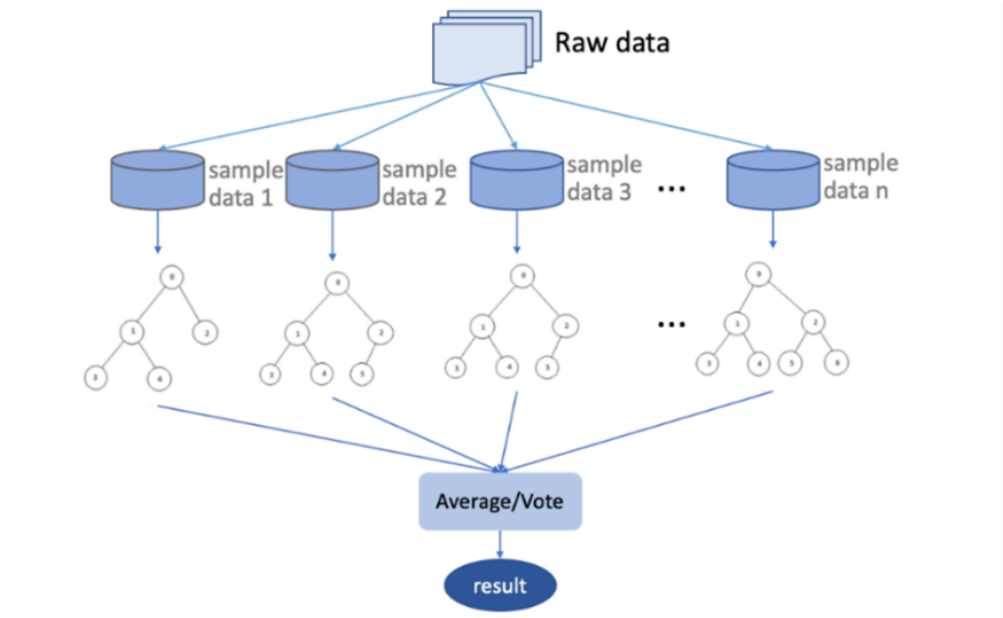


Figure 5.1 Random Forest Prediction Process

5.4 Match Advantage Prediction Model Results

Random Forest model is selected as the base model of the match advantage prediction model. We also select the features of 5.2.1 as the input of the model and the player advantage as the output of the model. The parameters of the model are optimized, and the parameters of the final optimal model were set as follows.

Table 5.3 Match advantage prediction model parameter setting

Parameters	Value	Parameters	Value
criterion	entropy	max_depth	10
bootstrap	False	max_feature	10

Then we get the results of the confusion matrix and evaluation index of the match advantage prediction model as follows.

Table 5.4 confusion matrix

	0	1	2
0	332	79	68
1	82	230	10
2	93	6	193

Table 5.5 Evaluation index result

	precision	recall	f1-score	support
0	0.65	0.69	0.67	479
1	0.73	0.71	0.72	322
2	0.71	0.66	0.69	292
accuracy			0.69	1093
macro avg	0.70	0.69	0.69	1093
weighted avg	0.69	0.69	0.69	1093

From the Table 5.5 and Table 5.6, no matter the match advantage is 0, 1, or 2. The value of each evaluation index is above 0.69, which means the model has a good result for identifying the player advantage in the current match situation.

5.5 Analysis of The Most Relevant Factors

5.5.1 Random Forest feature importance score

Based on the built Random Forest model, the importance score of each feature can be calculated by calculating the number of times each feature is used in the decision tree or the importance measure used for splitting. Common calculation methods include node purity-based and out-of-bag error-based methods.

1. The node purity based side is used to calculate variable importance scores based on the degree of improvement in the purity of the node splits. It measures how much each feature improves the purity of the model.
2. Out-of-Bag Error (OBE) based approach is where every decision tree, random forest model is constructed using a portion of the training samples and the remaining unused samples are known as out-of-bag samples. By making predictions on the out-of-bag samples and calculating the error, the contribution of each feature to the predictive power of the model can be evaluated.

5.5.2 Visual presentation of the most relevant factors

From 5.5.1 Random Forest's feature importance scores, we choose to compute the importance scores of each feature in the Match Advantage Prediction Model based on Out-of-Bag Error (OBE) based approach as the size of the correlation coefficient affecting the player's advantage. I.e., the larger the importance score of a feature, the higher the factor affects the player's advantage. After sorting the different feature importance scores, the indicators with importance scores greater than or equal to 0.05 were selected for visualization and analysis, as shown below.

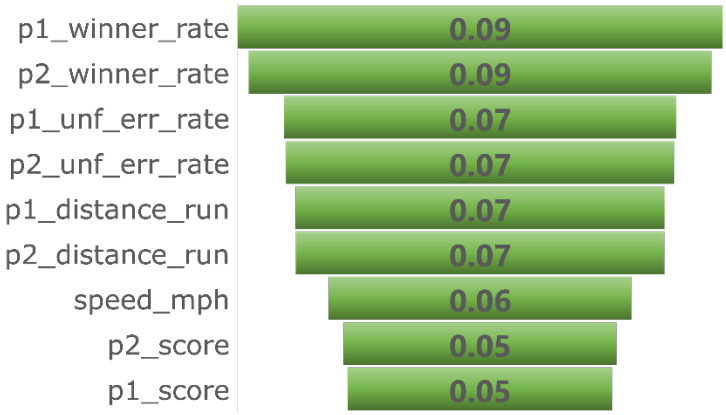


Figure 5.2 Demonstration of Factors Most Relevant to Player Advantage

From Figure 5.2， factors like p1_winner_rate,p2_winner_rate,p1_unf_err_rate，p2_unf_err_rate，p1_distance_run and p2_distance_run have higher importance scores, which means that these characteristics are of high importance in predicting player dominance. The possible reasons are as follows

1. p1_winner_rate,and p2_winner_rate denote the winning shot ratio of player1 and player2, and if a player's winning shot ratio continues to increase, then it will have a higher impact on the other player's mental condition, which will bring him or her an advantage in the game.
2. p1_unf_err_rate and p2_unf_err_rate indicate the percentage of forced error for player1

and player2. If a player has a larger percentage of forced error, that player is currently playing worse, reducing his or her advantage in the game.

3. p1_distance_run，p2_distance_run and speed_mph represent the running distance and bat speed of player1 and player2. They are reflection of the player's level of fitness and vigor. Higher running distances may imply that the players have better fitness and vigor, while lower running distances may imply that the players have low stamina or limited range of motion, which may have some impact on the match advantage.
4. p1_score and p2_score represent the current score situation of player1 and player2, and when the score is tied, it will have a certain impact on the psychological quality of the two players, and consequently on the game's advantages.

5.6 Advice on A Player Going into A New Match against A Different Player

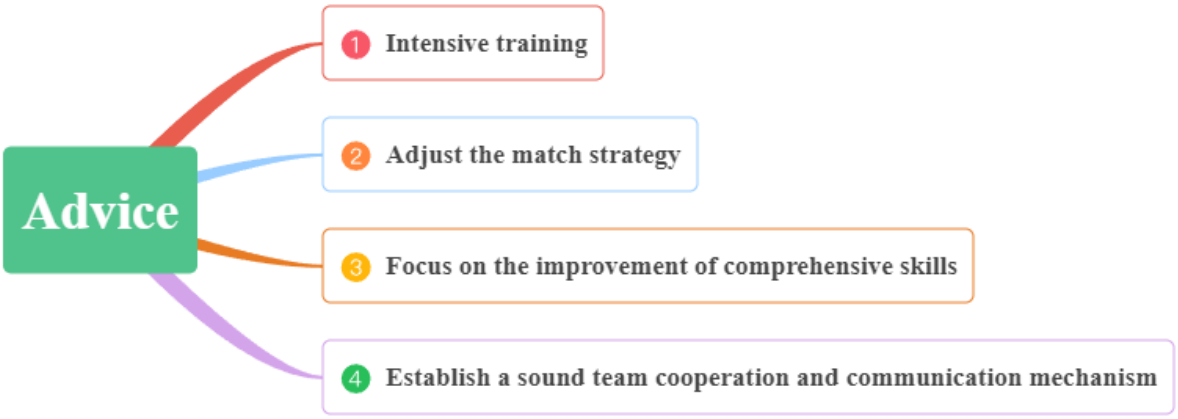


Figure 5.3 Advice for player how to against a different player

Advice 1: Intensive training. Analyzing players' transition point on momentum in the past matches can recognize whether there is a specific mode. Observe that whether players are more likely to change momentum at certain points. For instance, whether there is a tendency to relax when leading, or whether it is more aggressive when lagging behind. Based on the results of the analysis, the player's performance at possible turning points can be strengthened in a targeted manner. Improve the quality of serve when it comes to key points and strengthen mental preparation to improve stability and performance at key moments.

Advice 2: Adjust the match strategy. Compared to the data of the future opponents, it is possible to see if there are similar patterns of momentum shifts or vulnerabilities in some way. According to the opponent's vulnerable points and your own strengths, you can flexibly adjust the strategy. For example, when the opponent's physical decline or psychological in-stability, you can increase the attack, using the opponent's weakness to gain an advantage.

Advice 3: Focus on the improvement of comprehensive skills. Apart from stepping up their performances at possible turning points, players can focus on improving their overall skills, which includes aspects such as skills, tactical understanding and physical fitness. Through continuous technical training, tactical research and physical training, players can improve their overall level and thus better cope with various situations and dynamic changes in

the game.

Advice 4: Establish a sound team cooperation and communication mechanism. Cooperation and communication are the key factors of success. Players should establish a good communication mechanism with teammates, including tactical coordination, position adjustment and information exchange in the game. By strengthening teamwork and communication, players can better collaborate to respond to changes in opponents and develop more effective game strategies.

5.7 Test and Application on Match Advantage Prediction Model

First, we collected data for the remaining matches from the following websites.

Table 5.6 Competition data collection site

Data	Websites
2023-wimbledon-data	https://www.itftennis.com/en/
ping-pong match data	https://olympics.com/zh/olympic-games/tokyo-2020
2023-usopen-data	https://www.usopen.org/

5.7.1 Test on Match Advantage Prediction Model

We used the five matches from the additional collection of 2023 Wimbledon Gentlemen Championships to make predictions by using the match advantage prediction model we built, which included male tennis players like Carlos Alcaraz and Jeremy Chardy, and the final results are shown below.

Table 5.7 confusion matrix

	0	1	2
0	329	96	92
1	108	181	14
2	74	19	204

Table 5.8 Evaluation index result

	precision	recall	f1-score	support
0	0.64	0.64	0.64	517
1	0.61	0.60	0.60	303
2	0.66	0.69	0.67	297
accuracy			0.64	1117
macro avg	0.64	0.64	0.64	1117
weighted avg	0.64	0.64	0.64	1117

On the Table 5.7 and Table 5.8, the value of every evaluation index is above 0.64. Thus, our established Match Advantage Prediction Model has an excellent results in predicting player advantage in other matches of Wimbledon Gentlemen Championships.

Factors to be added in the future： Since our Match Advantage Prediction Model does not take into account players' physical fitness, skill levels, tactical strategies and so on. We only take into account changes in the form of the current match under the assumption that all of these are consistent, so that if the model performs poorly at times, indicators like players' physical fitness, skill levels, tactical strategies should be added to our model in the future.

5.7.2 Application on Match Advantage Prediction Model

1. 2023 Wimbledon Ladies Championships

We used the five matches from the additional collection of 2023 Wimbledon Ladies Championships to make predictions using the Match Advantage Prediction Model we built,

which included female tennis players like Ons Jabeur and Aryna Sabalenka, and the final predictions are shown below.

Table 5.9 confusion matrix

	0	1	2
0	260	65	57
1	67	125	5
2	78	5	112

Table 5.10 Evaluation index result

	precision	recall	f1-score	support
0	0.64	0.68	0.66	382
1	0.64	0.63	0.64	197
2	0.64	0.57	0.61	195
accuracy			0.64	774
macro avg	0.64	0.63	0.64	774
weighted avg	0.64	0.64	0.64	774

On the Table 5.9 and Table 5.10 the value of every evaluation index is above 0.64. Thus, our established Match Advantage Prediction Model has an excellent results in predicting player advantage in other matches of Wimbledon Ladies Championships too. Compared to the Table 5.7 and Table 5.8, we found that there is no significant difference when we use the model to predict players advantage in woman's match, **which illustrates that gender do not affect the applicability of the model.**

2. 2023 US Open Tennis Championships

We used our established Match Advantage Prediction Model to make predictions through an additional collection of three matches from the 2023 US Open Tennis Championships, which included tennis players like Alexander Zverev and Gijs Brouwer, and the final results are shown below.

Table 5.11 confusion matrix

	0	1	2
0	234	73	39
1	67	113	2
2	113	5	133

Table 5.12 Evaluation index result

	precision	recall	f1-score	support
0	0.62	0.68	0.65	346
1	0.59	0.62	0.61	182
2	0.76	0.62	0.68	215
accuracy			0.65	743
macro avg	0.66	0.63	0.65	743
weighted avg	0.65	0.64	0.65	743

On the Table 5.11 and Table 5.12, the value of every evaluation index is above 0.64. Therefore, our established Match Advantage Prediction Model has an excellent result in predicting player advantage in US Open Tennis Championships. Compared to Table 5.7 and Table 5.8, we discover that there is no significant difference when we use the model to predict players advantage in other tennis match, **which illustrates that the type of tennis match does not affect the applicability of the model.**

3. Table tennis competition at the Tokyo 2020 Olympics

We use our established Match Advantage Prediction Model to make predictions through an additional collection of Table tennis competition at the Tokyo 2020 Olympics between Sun Yinsha and Chen Meng, However, we only collectedame_no,set_no,p1_games,p2_games, p1_score,p2_score,server and point_victor features for the current match. Due to the limited

number of features we are able to collect, we apply equation (6) for calculating the momentum of Sun Yinsha and Chen Meng.

At the same time, retraining our established Match Advantage Prediction Model is necessary. Currently, we only select p1_games,p2_games,p1_score,p2_score features for training, and then, we use the trained model to predict the advantage of players in table tennis matches, and the final prediction results are shown below.

Table 5.13 confusion matrix

	0	1	2
0	58	3	8
1	15	0	0
2	10	0	14

Table 5.14 Evaluation index result

	precision	recall	f1-score	support
0	0.70	0.84	0.76	69
1	0.00	0.00	0.00	15
2	0.64	0.58	0.61	24
accuracy			0.67	108
macro avg	0.45	0.47	0.46	108
weighted avg	0.59	0.67	0.62	108

From Table 5.13 and Table 5.14, the model has a serious error in predicting player1 advantage, however, its accuracy can reach more than 0.5 in predicting player2 advantage, which shows that the model we built is also able to predict the player advantage of the rest of the ball game tournaments to a certain extent. **When the number of features for the remaining ball sport events is enough, we are certain that our model is also able to predict their player dominance outcomes better.**

6 Sensitivity Analysis

In order to determine the sensitivity of our model to the Match Advantage, we perform the following sensitivity analysis. First, according to Figure 5.2, it can be concluded that the greatest degree of importance is the winner_rate, however, the rate variance is difficult to change. Thus, to execute sensitivity analysis, we choose the distance_run variance to change while keeping the rest of the features constant. The specific steps are as follows:

Step 1. Divide the underlying data set into 85% training set and 15% test set.

Step 2. Use the Match Advantage Prediction Model for training and testing.

Step 3. Perturb the p1_distance_run feature and p1_distance_run feature of the 80% training set by 10%, 20% and 30%, respectively, and repeat **Step 2**.

The final test set fit obtained as shown in the Table 6.1.

Table 6.1Results of sensitivity change analysis

perturbation values of variables	accuracy	precision	recall	f1-score
$\alpha_{distance_run} = 0\%$	0.69	0.69	0.69	0.69
$\alpha_{distance_run} = 10\%$	0.69	0.69	0.69	0.69
$\alpha_{distance_run} = 20\%$	0.68	0.68	0.68	0.68
$\alpha_{distance_run} = 30\%$	0.69	0.69	0.69	0.69

From the Table 6.1, when perturbing distance_run, the variation of the fitting effect is small and within an acceptable range, thus our proposed Match Advantage Prediction Model has a fairly good robustness in predicting Match Advantage.

7 Strengths and Weaknesses

7.1 Strengths

- **Intuitive visualization:** by breaking down the game into each point, game and set and visualizing them, we can intuitively observe and compare the performance of the players.
- **Good robustness:** we conduct a sensitivity analysis of the model, and our model still maintains a relatively stable output when there are small changes, indicating that the model has good robustness.
- **Good generalizability:** the model can adapt to different players' skill levels and make suggestions based on their needs, and it can also be migrated to different sports, such as table tennis.

7.2 Weaknesses

- Limitations: due to the lack of data, the model only analyzes based on the active level of the players and does not take into account other important factors (e.g., the player's physical quality, technical level, tactical strategy, etc.).
- Lack of dynamism: the model is based on static data analysis and cannot capture the dynamic changes during the game.

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Report on Use of AI

OpenAI ChatGPT (Feb, 2024 version, ChatGPT-3.5)

Query1: Could you please help me solve some problems in the contest?

Output1: Yes. In tennis, the concept of momentum refers to the psychological and physiological advantage gained by a player or team when they are on a winning streak or have the momentum of the match going in their favor. It is often described as a feeling of confidence, energy, and control. When a player has momentum, they tend to perform better, make fewer errors, and their opponent may feel more pressure.

Query2: I want to draw a poster about tennis. What information does the poster need to include?

Output2: To create a poster about tennis, include the following information:

Title: Use an attention-grabbing and descriptive title.

Imagery: Include visually appealing images related to tennis.

Key Information: Provide a brief overview of tennis, including basic rules and scoring.

Tournament or Event Details: If applicable, mention the name, date, and location of a specific tournament or event.

Benefits: Highlight the advantages of participating in or watching tennis, such as physical fitness and social interaction.

Query3: What factors are the performance of tennis players mainly related to?

Output3: The performance of tennis players is mainly related to the following factors:

Technical Skills: Proficiency in strokes, footwork, serve, and other fundamental tennis skills.

Physical Fitness: Strength, speed, agility, endurance, flexibility, and coordination.

Mental Strength: Focus, concentration, resilience, confidence, and handling pressure.

Strategy and Tactics: Equationting and executing effective game plans.

Experience and Match Play: Exposure to different opponents and match situations.

Appendix

Task1: Momentum Calculation

```
def calculate_momentum(data):
    x1, x2 = [], []
    p1_momentum = 0
    p2_momentum = 0
    consecutive_score = 0

    previous_point_victor = None
    for i in range(len(data)):
        point_victor = data.loc[i, 'point_victor']
        server = data.loc[i, 'server']
        p1_break_pt_won = data.loc[i, 'p1_break_pt_won']
        p2_break_pt_won = data.loc[i, 'p2_break_pt_won']
        p1_unf_err = data.loc[i, 'p1_unf_err']
        p2_unf_err = data.loc[i, 'p2_unf_err']

        if point_victor == 1:
            p1_momentum += 1
            p2_momentum = 0
        elif point_victor == 2:
            p1_momentum = 0
            p2_momentum += 1

        if point_victor == 1:
            weight1 = 0.8 if server == 1 else 1.2
            weight2 = 1
        elif point_victor == 2:
            weight1 = 1
            weight2 = 0.8 if server == 2 else 1.2

        p1_momentum = p1_momentum * weight1
        p2_momentum = p2_momentum * weight2

        if point_victor == previous_point_victor:
            consecutive_score += 1
            if consecutive_score >= 3:
                p1_momentum += 5 if point_victor == 1 else 0
                p2_momentum += 5 if point_victor == 2 else 0
        else:
            consecutive_score = 1

        x1.append(p1_momentum + p1_break_pt_won * 5 - p1_unf_err * 5)
        x2.append(p2_momentum + p2_break_pt_won * 5 - p2_unf_err * 5)

        previous_point_victor = point_victor

    data['p1_momentum'] = x1
    data['p2_momentum'] = x2

        return data
```